Machine Learning Model For Flood Prediction

Abstract

Climate change has significantly increased the occurrence and severity of floods, posing serious threats to human lives, infrastructure, agriculture, and the environment. As one of the major climate-related disasters, floods require accurate and timely prediction to minimize their impact. This project presents a Machine Learning-based Flood Prediction System designed to support Sustainable Development Goal (SDG) 13 – Climate Action. The proposed model uses historical weather and environmental data, including rainfall, temperature, humidity, river water level, soil moisture, and atmospheric pressure, to predict the likelihood of flood occurrence. Data preprocessing techniques such as cleaning, normalization, and feature selection are applied to improve the quality of the dataset before training the model. Among various machine learning algorithms, the Random Forest algorithm is selected due to its high prediction accuracy, robustness, and ability to handle complex environmental data. The trained model can provide early flood warnings, enabling governments, disaster management agencies, and local communities to take preventive measures and reduce potential damage. By improving disaster preparedness and strengthening resilience against climate-related hazards, this project contributes directly to the objectives of SDG 13 – Climate Action, promoting sustainable development and enhancing community safety through intelligent decision-making.

Problem Statement

Climate change has led to an increase in the frequency and intensity of extreme weather events, making floods one of the most common and destructive natural disasters worldwide. Heavy rainfall, rising river levels, and changing weather patterns often result in severe flooding, causing loss of life, damage to infrastructure, disruption of agriculture, and significant economic losses. Traditional flood forecasting methods may not always provide accurate or timely predictions due to the complexity of environmental factors.

To address this challenge, there is a need for an intelligent and data-driven approach that can analyze historical climate and environmental data to predict flood occurrences with greater accuracy. This project proposes a Machine Learning-based Flood Prediction System that utilizes features such as rainfall, temperature, humidity, river water level, soil moisture, and atmospheric pressure to predict the likelihood of flooding. By providing early warnings and supporting informed decision-making, the proposed model contributes to Sustainable Development Goal (SDG) 13 – Climate Action by improving disaster preparedness, reducing flood-related risks, and strengthening community resilience against climate-related hazards.

Objectives

The primary objective of this project is to design a Machine Learning-based Flood Prediction System that supports Sustainable Development Goal (SDG) 13 – Climate Action by providing accurate and timely flood predictions using historical weather and environmental data.

Specific Objectives

1. To develop a machine learning model that predicts the likelihood of flood occurrence based on climate and hydrological parameters.
2. To collect and preprocess historical weather and environmental data, including rainfall, temperature, humidity, river water level, and soil moisture, for effective model training.
3. To implement the Random Forest algorithm for accurate and reliable flood prediction.
4. To evaluate the model's performance using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
5. To provide an early warning system that assists disaster management authorities and local communities in taking preventive actions.
6. To support SDG 13 – Climate Action by enhancing disaster preparedness, reducing the impact of floods, and improving climate resilience through intelligent prediction.

Machine Learning Algorithm

1. Algorithm Used

- Random Forest (Supervised Machine Learning Algorithm)
- Used for binary classification (Flood / No Flood).
- Combines multiple Decision Trees for better prediction accuracy.

2. How It Works

- Collect and preprocess historical climate data.
- Train multiple Decision Trees using random subsets of the data.
- Each tree predicts Flood or No Flood.
- Final prediction is made using majority voting.

3. Why Random Forest?

- High prediction accuracy.
- Reduces overfitting compared to a single Decision Tree.
- Handles multiple climate features and non-linear data effectively.
- Provides feature importance to identify key flood-causing factors.

4. Other Algorithms That Can Be Used

- Decision Tree – Simple and easy to understand but prone to overfitting.
- Logistic Regression – Suitable for simple binary classification but less accurate for complex data.
- Support Vector Machine (SVM) – Good accuracy for small datasets but slower for large datasets.
- XGBoost – Very high accuracy but requires more computation and parameter tuning.

Comparison of Algorithms

Algorithm	Accuracy	Overfitting	Training Speed	Suitable for Flood Prediction
Logistic Regression	Medium	Low	Fast	Moderate
Decision Tree	Medium	High	Fast	Good
Support Vector Machine	High	Low	Slow (Large Data)	Good
XGBoost	Very High	Low	Moderate	Excellent
Random Forest	**High**	**Very Low**	**Fast**	**Excellent (Selected)**

5. Why Random Forest is Better?

- More accurate and reliable than Decision Tree and Logistic Regression.
- Less prone to overfitting due to ensemble learning.
- Works well with large datasets and multiple environmental factors.
- Easy to implement while providing strong performance, making it ideal for flood prediction under SDG 13 – Climate Action.

Methodology

1) **Data Collection**

   Collect historical climate and environmental data such as rainfall, temperature, humidity, river water level, and soil moisture.

2) **Data Preprocessing**

   Clean the dataset by removing missing values and duplicates, normalize the data, and select relevant features.

3) **Model Training**

   Split the dataset into training and testing sets, then train the Random Forest algorithm using the training data.

4) **Prediction and Evaluation**

   Test the model on unseen data and evaluate its performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

5) **Flood Risk Prediction**

   Use the trained model to predict flood occurrence and generate early warnings, supporting SDG 13 – Climate Action by improving disaster preparedness.

SAMPLE INPUT:

If the risk of flood is high the sample input will be :

Feature	Value
Rainfall	280 mm
Temperature	28 °C
Humidity	92 %
River Level	8.5 m
Soil Moisture	88 %
Pressure	995 hPa
Wind Speed	18 km/h

SAMPLE OUTPUT:

```
Flood Probability : 97.45%

Prediction : HIGH FLOOD RISK
```

INPUT:

```python
# ========================================
# FLOOD PREDICTION USING RANDOM FOREST
# SDG 13 - Climate Action
# ========================================

# Install libraries (if needed)
!pip -q install seaborn

# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)

# ========================================
# Upload Dataset
# ========================================

print("Upload Flood_Prediction.csv")

uploaded = files.upload()
```

```python
print("\nClassification Report\n")

print(classification_report(y_test, y_pred))

# ========================================
# Confusion Matrix
# ========================================

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,5))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['No Flood','Flood'],
    yticklabels=['No Flood','Flood']
)

plt.title("Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.show()
```

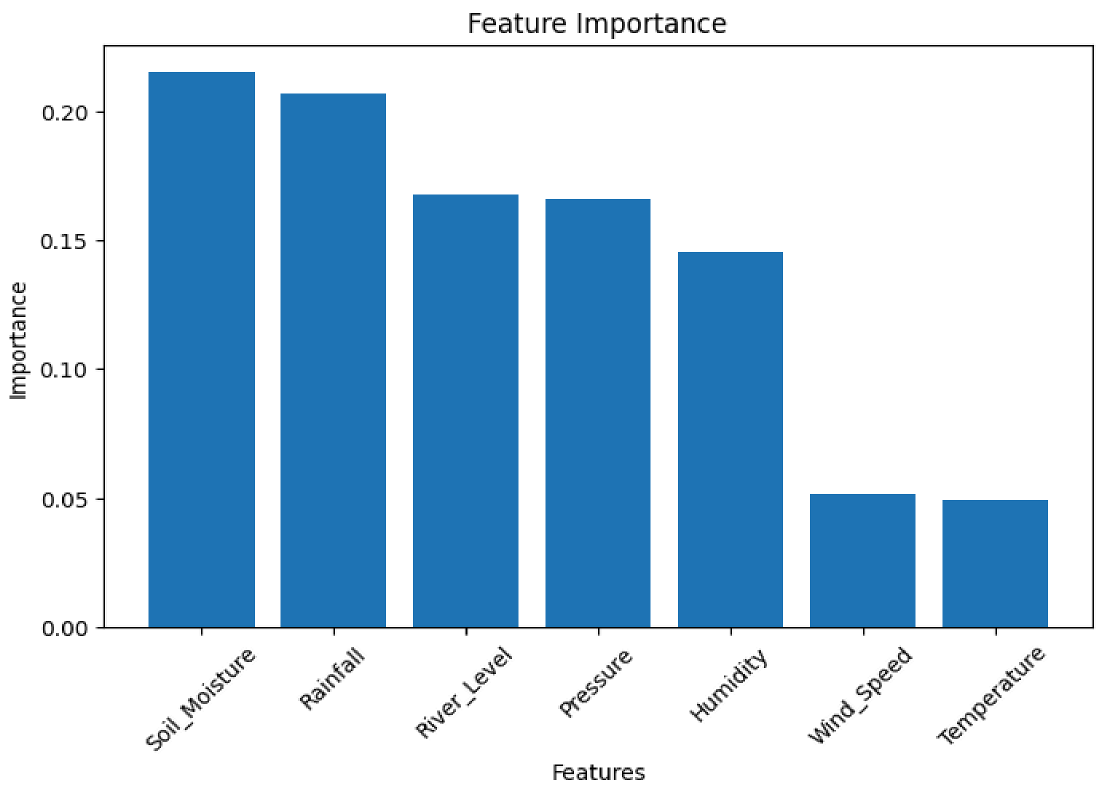
OUTPUT:

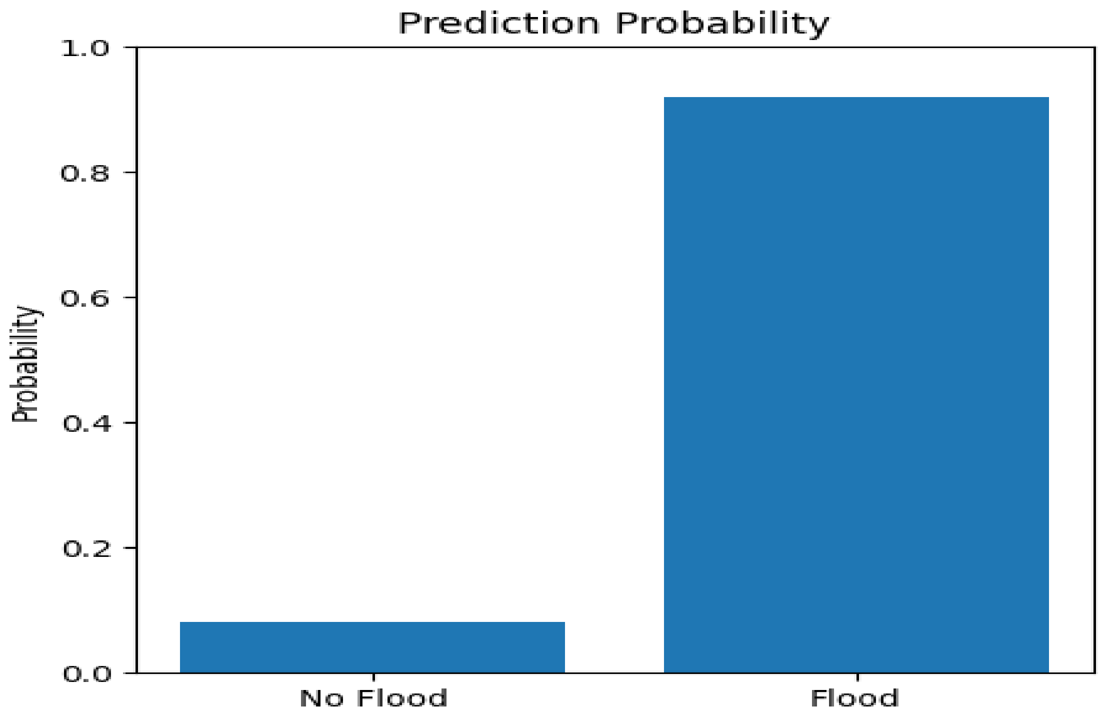
```
Dataset Preview
   Rainfall  Temperature  Humidity  River_Level  Soil_Moisture  Pressure  \
0       327           21        33          7.7             51       994
1       377           21        99          1.8             74       982
2        47           24        59          5.5             23      1015
3       366           38        99          4.8             77      1017
4         3           23        84          4.1             39       993

   Wind_Speed  Flood
0           8      1
1           1      1
2          12      0
3          17      1
4          21      0

Dataset Shape: (500, 8)
```

```
===================================
MODEL ACCURACY
===================================
Accuracy : 96.00%

Classification Report

              precision    recall  f1-score   support

           0       0.97      0.99      0.98        84
           1       0.93      0.81      0.87        16

    accuracy                           0.96       100
   macro avg       0.95      0.90      0.92       100
weighted avg       0.96      0.96      0.96       100
```

```
========== NEW PREDICTION ==========

Flood Probability : 92.00%

Prediction : HIGH FLOOD RISK
```

GRARH:


Feature Importance
0.20
0.15
0.10
0.05
0.00
Importance
Soil_Moisture
Rainfall
River_Level
Pressure
Humidity
Wind_Speed
Temperature
Features


Prediction Probability
1.0
0.8
0.6
0.4
0.2
0.0
Probability
No Flood
Flood

Conclusion

This project successfully demonstrates the design of a Machine Learning-based Flood Prediction System using the Random Forest algorithm. By analyzing climate and environmental factors such as rainfall, temperature, humidity, river water level, soil moisture, pressure, and wind speed, the model can accurately predict the likelihood of flooding. The proposed system supports SDG 13 – Climate Action by enabling early flood warnings, improving disaster preparedness, and reducing the impact of climate-related hazards. Overall, this project highlights the potential of machine learning in building smarter, data-driven solutions for environmental protection and sustainable development.